

Software Engineer Trial Project (React / Relay)

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Overview

This is a simple project that will showcase your abilities and provide a starting point for some technical discussion during the interview process. It's intended to be short, not requiring more than two hours of your time, but flexible if you are interested in putting in more time.

Project

We ask that you build a front end for the Random User GraphQL service, taking a couple of parameters and displaying some statistics about the random users that are fetched. The front end must be written in TypeScript using React and Relay.

The statistics you will calculate for the users are:

- Percentage of users by gender
- Percentage of users in the following age ranges: 0-20, 21-40, 41-60, 61-80, 81-100, 100+
- Counts of users by length of last name
- Percentage of users in each state for the 10 most common states in the results

You can display this information as simple text.

Your front end must include two input fields; the exact UX is up to you:

- the number of users to fetch from the GraphQL endpoint, defaulting to 200
- the nationality of users to restrict the results to, defaulting to `us`

When the user changes the input parameters, they must be validated, and if they are valid, a new list of users must be fetched from the GraphQL endpoint, and the statistics recalculated accordingly.

See below for details on the GraphQL service, what these values mean, and what values are acceptable.

GraphQL service

The GraphQL service's *GraphiQL* interface can be found here:

<https://nextjs-randomuser-graphql.vercel.app/api/graphql>

Here's a simple curl request to demonstrate how it works:

```
curl 'https://nextjs-randomuser-graphql.vercel.app/api/graphql' -H 'content-type: application/json' \
  --data-raw '{"query":"{users(results: 5) {name {last}}}"}
```

A more full-featured query looks something like this:

```
query usersByNationality($nat: String!) {
  users(
    nat: $nat
    results: 20
  ) {
    name {
      first
      last
    }
    gender
    dob {
      date
      age
    }
  }
}
```

You will need to create your own query to get the information and filter appropriately for this project.

The relevant filters for this project are:

- `nat` — a nationality, which is basically a two-letter country code from the list here: <https://randomuser.me/documentation#nationalities> (any version of the list is acceptable)
- `results` — an integer from 1 to 5000 indicating how many users to fetch

Deliverables

We will run your project ourselves to check the functionality. Please send to us the source code to your project and documentation on how to run it. You may send us a link to a hosted source repository (for example at GitHub), or you can send us an archive file (zip, tar, or similar) with the contents. If you send an archive file, please include your `.git` directory, as we are interested in the development history of your project, too.

If you have any questions about the project, please let us know as soon as possible.

Extras

If you enjoy working on the project and feel like spending some more time with it, please feel free to expand upon it, so long as the basic requirements are still met. For example, you could:

- add pie charts or histograms for the statistics
- include other statistics that you find interesting
- create other inputs to filter the results